SF1811 Optimization
Formula sheet for final exam 2026/2027

Note: No calculator is allowed on the exam!

- **Four fundamental subspaces.**
  $\mathcal{R}(A)^\perp = \mathcal{N}(A^T),\ \mathcal{R}(A^T)^\perp = \mathcal{N}(A),\ \mathcal{N}(A)^\perp = \mathcal{R}(A^T),\ \mathcal{N}(A^T)^\perp = \mathcal{R}(A)$.

- **Simplex method for LP problem in standard form.**
  $A = (A_B\ A_N)$, $A_B x_B = b$, $z = c_B^T x_B$, $A_B^T y = c_B$, $r_N = c_N - A_N^T y$. Finished if $r_N \geq 0$. Otherwise, choose $k \in N$ with $r_k < 0$. $x_k = t$, $A_B \Delta x_B = -a_k$, $z(t) = c_B^T x_B + r_k t$, $x_B(t) = x_B + \Delta x_B t$, $t_{\max} = \min\limits_{i:(\Delta x_B)_i<0} \dfrac{(x_B)_i}{-(\Delta x_B)_i} = \dfrac{(x_B)_p}{-(\Delta x_B)_p}$. Let k and B_p change place.

- **Optimality conditions for LP problem in general form.**

$$(P)\quad \begin{array}{ll} \text{minimize} & c_1^T x_1 + c_2^T x_2 \\ \text{subject to} & A_{11}x_1 + A_{12}x_2 \geq b_1, \\ & A_{21}x_1 + A_{22}x_2 = b_2, \\ & x_1 \geq 0,\ x_2 \text{ free.} \end{array} \qquad (D)\quad \begin{array}{ll} \text{maximize} & b_1^T y_1 + b_2^T y_2 \\ \text{subject to} & A_{11}^T y_1 + A_{21}^T y_2 \leq c_1, \\ & A_{12}^T y_1 + A_{22}^T y_2 = c_2, \\ & y_1 \geq 0,\ y_2 \text{ free.} \end{array}$$

  $\hat{x}$ is optimal to (P) and $\hat{y}$ is optimal to (D) if and only if $\hat{x}$ is feasible to (P), $\hat{y}$ is feasible to (D), $\hat{y}_1^T(A_{11}\hat{x}_1 + A_{12}\hat{x}_2 - b_1) = 0$ and $\hat{x}_1^T(c_1 - A_{11}^T\hat{y}_1 - A_{21}^T\hat{y}_2) = 0$.

- **Factorization of symmetric positive [semi]definite matrix.** A symmetric matrix H is positive definite [semidefinite] if and only if there is a lower triangular matrix L with all $l_{ii} = 1$ and a diagonal matrix D with all $d_i > 0$ [$d_i \geq 0$] such that $H = LDL^T$.

- **Quadratic functions.** $f(x) = \frac{1}{2}x^T H x + c^T x + c_0$, with H symmetric.
  $\nabla f(x) = (Hx + c)^T$, $F(x) = H$, $f(x + td) = f(x) + t(Hx + c)^T d + \frac{1}{2}t^2 d^T H d$.
  $\hat{x}$ minimizes $f(x)$ if and only if H is positive semidefinite and $H\hat{x} + c = 0$.

- **Equality-constrained QP.** minimize $\frac{1}{2}x^T H x + c^T x + c_0$ subject to $Ax = b$.
  $\hat{x} = \bar{x} + Z\hat{v}$, $\quad (Z^T H Z)\hat{v} = -Z^T(H\bar{x} + c)$, $\quad \begin{pmatrix} H & -A^T \\ A & 0 \end{pmatrix}\begin{pmatrix} \hat{x} \\ \hat{u} \end{pmatrix} = \begin{pmatrix} -c \\ b \end{pmatrix}$.

- **MN solution to LSQ problems.** $A^T A\bar{x} = A^T b$, $\bar{x} = \hat{x}_{\text{MN}} + z$, $\hat{x}_{\text{MN}} \in \mathcal{R}(A^T)$, $z \in \mathcal{N}(A)$.

- **Newton.** $F(x^{(k)})d^{(k)} = -\nabla f(x^{(k)})^T$ gives $d^{(k)}$ if $F(x^{(k)})$ is positive definite.
  $x^{(k+1)} = x^{(k)} + t_k d^{(k)}$ where t_k satisfies $f(x^{(k)} + t_k d^{(k)}) < f(x^{(k)})$. Try $t_k = 1$ first.

- **Nonlinear LSQ.** minimize $f(x) = \frac{1}{2}\sum_{i=1}^m (h_i(x))^2 = \frac{1}{2}h(x)^T h(x)$.
  $x \in \mathbb{R}^n$, $h(x) \in \mathbb{R}^m$, $\nabla h(x)$ an $m \times n$ matrix with $\dfrac{\partial h_i}{\partial x_j}(x)$ in row i and column j.
  $\nabla f(x) = h(x)^T \nabla h(x)$ (row vector), $F(x) = \nabla h(x)^T \nabla h(x) + \sum_{i=1}^m h_i(x) H_i(x)$.

- **Gauss-Newton**. $\nabla h(x^{(k)})^T \nabla h(x^{(k)}) d^{(k)} = -\nabla h(x^{(k)})^T h(x^{(k)})$ gives $d^{(k)}$.
  $x^{(k+1)} = x^{(k)} + t_k d^{(k)}$ where t_k satisfies $f(x^{(k)} + t_k d^{(k)}) < f(x^{(k)})$. Try $t_k = 1$ first.

- **Constrained NLP.** minimize $f(x)$ subject to $h_i(x) = 0$, $i \in \mathcal{E}$, $g_i(x) \leq 0$, $i \in \mathcal{I}$.
  KKT conditions:
  $\nabla f(\hat{x}) + \sum_{i\in\mathcal{E}} \hat{u}_i \nabla h_i(\hat{x}) + \sum_{i\in\mathcal{I}} \hat{y}_i \nabla g_i(\hat{x}) = 0^T$,
  $h_i(\hat{x}) = 0$, $i \in \mathcal{E}$, $g_i(\hat{x}) \leq 0$, $i \in \mathcal{I}$, $\hat{y}_i \geq 0$, $i \in \mathcal{I}$, $\hat{y}_i g_i(\hat{x}) = 0$, $i \in \mathcal{I}$.

- **Lagrangian relaxation.** (P): minimize $f(x)$ subject to $h(x) = 0$, $g(x) \leq 0$ and $x \in X$.
  $L(x, u, y) = f(x) + u^T h(x) + y^T g(x)$, $\varphi(u, y) = \min_{x\in X} L(x, u, y)$.
  (D): maximize $\varphi(u, y)$ subject to $y \geq 0$.
  Glob. opt. cond. (GOC): $L(\hat{x}, \hat{u}, \hat{y}) = \min\limits_{x\in X} L(x, \hat{u}, \hat{y})$, $h(\hat{x}) = 0$, $g(\hat{x}) \leq 0$, $\hat{y} \geq 0$, $\hat{y}^T g(\hat{x}) = 0$.